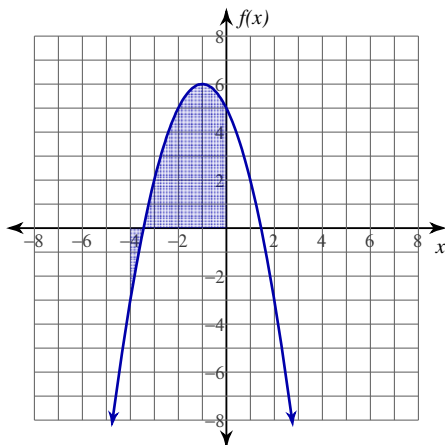


Mean Value Theorem for Integrals

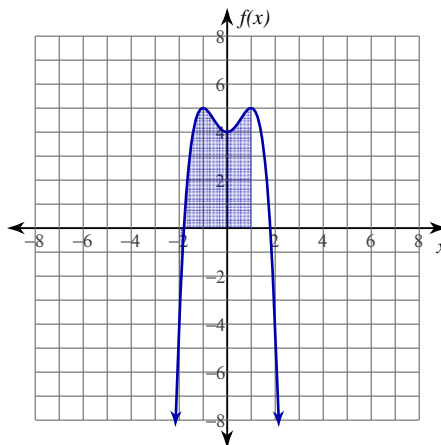
Date _____ Period _____

For each problem, find the average value of the function over the given interval.

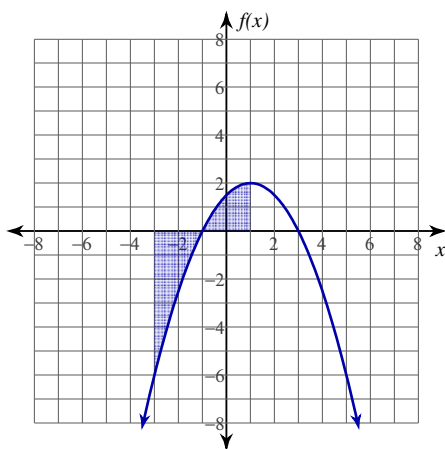
1) $f(x) = -x^2 - 2x + 5$; $[-4, 0]$



2) $f(x) = -x^4 + 2x^2 + 4$; $[-2, 1]$

**For each problem, find the values of c that satisfy the Mean Value Theorem for Integrals.**

3) $f(x) = -\frac{x^2}{2} + x + \frac{3}{2}$; $[-3, 1]$



4) $f(x) = \frac{4}{x^2}$; $[-4, -2]$

